

ance, etc could tap into this system and thereby work out strategies to better service the customers based on the way they handle their vehicles.

The system has capabilities to flag certain non-compliant events that could endanger the vehicle such as a faulty door/engine speed-limit infringement, and initiate emergency notifications. Critical minutes of an unfavourable event happening to the vehicle and the passengers are reduced in this process, reducing changes of accidents. The system is programmed to feed in emergency contacts so that help reaches the spot where the vehicle was last reported. This is done by combining software, hardware and Cloud connectivity powered by analytics. Media, connectivity skills and embedded skills, all contribute to designing this system.

The next step

Towards the end of the infotainment project, the teams involved asked themselves a rather simple and regular question: "How can these features be designed better to add more value to end users—how can we further improve the driving experience?"

Amidst all the discussions and research, the teams started toying with the idea of a more sophisticated display that also enhances safety. A return to the sketchboard showed that there were very few after-market head-up display (HUD) products, and most had a cluttered and user-unfriendly UI design.

Leveraging the experience of the infotainment project, Embitel developed a clutter-free and user-friendly hardware and software design of Car

The in-car infotainment system

- The system is designed on Freescale's i.MX6 platform and Android 4.2.2
- Stock Android was modified to suit IVI application
- Smartphone pairing has been done through Bluetooth profiles to access phone contacts and phone apps, receive/make calls through the dashboard, access phone messages and external multimedia/radio systems
- The system is integrated with other onboard electronic control units to fetch vehicle data for diagnostics
- This IVI system has instrument clusters and virtual gauges that fetch vehicle data through control area networks
- The project team integrated automotive-specific, third-party apps for navigation purposes and for locating nearest fuel stations
- This infotainment system supports voice commands to mitigate driver distraction

HUD. You can see the layout and design in Fig. 1. Although the final design looks simple, the process to come up with such a design involved a lot of work.

This proprietary design is production-ready and is of automotive-grade. This ensures that a customer can launch the Car HUD product in six months.

The HUD displays real-time driving data coupled with road signs and lane departure alerts as it continuously monitors the set safety parameters. The touch-less commands through their integrated gesture-control and voice-recognition modules are able to give a hands-free seamless experience without distractions of any kind.

Users can dictate their response, attend/reject calls using simple hand motions or ask the unit to read these via a voice assist. The microphone picks up the voice from the user and directs it to a voice-

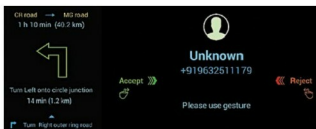


Fig. 2: UI of Car HUD

recognition unit than then processes the voice. Operations are performed once the particular command key words are identified. Commands are recognised and interpreted using the data dictionary to perform actions, which can be as simple as saying Accept or Reject (the call). The UI is equally responsive.

Heads up to the future

Embitel Technologies started its journey in 2006. Since then the product engineering services business has collaborated with global customers for R&D and embedded product development services in automotive and home/factory automation areas.

The cross-functional expertise in mobile and Cloud connectivity technologies has helped add value to the customer base with solutions for the Internet of Things (IoT) for home automation, healthcare, infotainment, telematics, advanced driver assistance systems and connected cars. It remains to be seen how the market evolves, now that Google is about to come up with their own solution! **EFY**

Here is what the product comes with

- The system runs on Android Kitkat 4.4 platform
- Real-time navigation and display on the windshield through PICO projectors
- Display screen on the windshield with anti-reflect for distraction-free viewing
- Voice recognition and gesture control to enable touch-less interfaces
- Wi-Fi cameras for all proximity capture visuals and dash cam to detect driver position
- BLE 4.0 interface for pairing smartphones and devices for managing calls, messages and media streaming
- Customisable UI for better user experience